Double integral

F18CE Week 9

Definition 8.1: Bounded region in $\mathbb{R}^2$

A set $D \subset \mathbb{R}^2$ is called bounded if $\exists R > 0$ such that $D \subset D_R = \left\{ \left(x, y \right) : x^2 + y^2 < R^2 \right\}$.

In other words, D is said to be bounded if it fits inside a sufficiently large disc.

Definition 8.2: Double integrals

We define the double integrals from generalising the concept of Riemann sum and Riemann integral in the previous Chapter. Check out the notes posted in Teams with title “Definition of double integral”.

Theorem 8.3: Integrability of functions

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function. Given a bounded region D on a plane. Suppose f is continuous on D (except for possibly finitely many discontinuities), then f is integrable on D.

Theorem 8.4: Properties of Double integrals

Suppose $f, g : \mathbb{R}^2 \to \mathbb{R}$ are integrable on a bounded region $D \subset \mathbb{R}^2$ and $C \in \mathbb{R}$ is a constant. Then

a) $\iint\limits_D Cf(x,y)dxdy = C\iint\limits_D f(x,y)dxdy$

b) $\iint\limits_D \Big(f(x,y) + g(x,y)\Big)dxdy = \iint\limits_D f(x,y)dxdy + \iint\limits_D g(x,y)dxdy$

c) $\iint\limits_D f(x,y)dxdy \leq \iint\limits_D g(x,y)dxdy$ if $f(x,y) \leq g(x,y)\ \forall (x,y) \in D$

Theorem 8.5: Properties of Double integrals

Suppose $D_1, D_2 \subset \mathbb{R}^2$ such that $D_1 \cap D_2 = \phi$ and $D = D_1 \cup D_2$

.

D_1 and D_2 are partition of D). Then for any function $f = f(x,y) : \mathbb{R}^2 \to \mathbb{R}$ which is integrable on D_1 and on D_2, f is integrable on D and

$$\iint_D f(x,y)dxdy = \iint_{D_1} f(x,y)dxdy + \iint_{D_2} f(x,y)dxdy\,.$$

Remark

Remark: In general, if f is integrable on $D_1, D_2, ..., D_n$ and $D_1, D_2, ..., D_n$ are partition of D, then f is integrable on D and

$$\iint_D f(x,y)dxdy = \iint_{D_1} f(x,y)dxdy + \iint_{D_2} f(x,y)dxdy + \cdots + \iint_{D_n} f(x,y)dxdy\,.$$

Theorem 8.6: Variables interchange in double integral

Let $R = \{(x, y) : a \leq x \leq b, c \leq y \leq d\}$ be a rectangle, and $f : \mathbb{R}^2 \to \mathbb{R}$ be a function integrable on R. Suppose $\int_a^b f(x, y)\, dx$ exists for all values of $y \in [c, d]$, then $\int_c^d \left(\int_a^b f(x, y)\, dx \right) dy$ exists, and $\iint_R f(x, y) dx dy = \int_c^d \left(\int_a^b f(x, y)\, dx \right) dy$.

Remark: The roles of x and y in the above theorem can be interchanged, i.e.,

$$\iint_R f(x, y) dx dy = \int_a^b \left(\int_c^d f(x, y)\, dy \right) dx .$$

Make use of this idea to solve the integration when necessary.

Example 8.7

Let $f(x,y) = 3xy^2 + 2x^3$ and $R = \{(x,y) : 1 \le x \le 2, 0 \le y \le 1\} \subset \mathbb{R}^2$.

$$\begin{aligned}
\iint_R f(x,y)dydx &= \int_1^2 \left(\int_0^1 (3xy^2 + 2x^3)\, dy \right) dx . \\
&= \int_1^2 \left(\left[xy^3 + 2x^3 y \right]_{y=0}^{y=1} \right) dx \\
&= \int_1^2 \left(x + 2x^3 \right) dx \\
&= \tfrac{1}{2} \left[x^2 + x^4 \right]_1^2 \\
&= 9
\end{aligned}$$

Or,

$$\begin{aligned}
\iint_R f(x,y)dxdy &= \int_0^1 \left(\int_1^2 (3xy^2 + 2x^3)\, dx \right) dy \\
&= \tfrac{1}{2} \int_0^1 \left(\left[3x^2y^2 + x^4 \right]_{x=1}^{x=2} \right) dy \\
&= \tfrac{1}{2} \int_0^1 \left(9y^2 + 15 \right) dy \\
&= \tfrac{1}{2} \left[3y^3 + 15y \right]_0^1 \\
&= 9
\end{aligned}$$

Theorem 8.8: Variables interchange (generalisation of Theorem 8.6)

Let D be a region bounded by continuous curves $y = \alpha(x)$ and $y = \beta(x) \geq \alpha(x)$ with $a \leq x \leq b$. Let $f(x, y) : \mathbb{R}^2 \to \mathbb{R}$ be a function integrable on D.

Suppose $\int_{y=\alpha(x)}^{y=\beta(x)} f(x,y)\,dy$ exists for all $x \in [a, b]$, then $\int_a^b \left(\int_{y=\alpha(x)}^{y=\beta(x)} f(x,y)\,dy \right) dx$ exists and

$$\iint_D f(x,y)dxdy = \int_a^b \left(\int_{y=\alpha(x)}^{y=\beta(x)} f(x,y)\,dy \right) dx\,.$$

Remark

Remark: Similarly, we can have $x = \gamma(y)$ and $x = \delta(y)$, with $c \leq y \leq d$, then

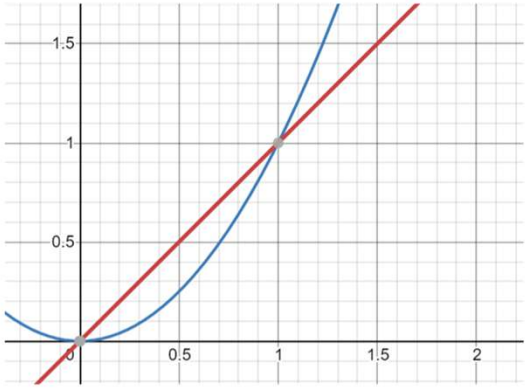
$$\iint_D f(x,y)dxdy = \int_c^d \left(\int_{x=\gamma(y)}^{x=\delta(y)} f(x,y)\,dx \right) dy\,.$$

Example 8.9

a) Determine $\iint_D xy\,dxdy$, where D is the area in the first quadrant between the curves $y = x$ and $y = x^2$.

Solution. We need to be careful when we compute. We need to ensure that $\beta(x) \geq \alpha(x)$ with the given range of x. We see that $0 \leq x \leq 1$ and $x^2 \leq y \leq x$. Hence,

$$\begin{aligned}
\iint_D xy\,dxdy &= \int_0^1 \left(\int_{y=x^2}^{y=x} xy\,dy \right) dx \\
&= \frac{1}{2} \int_0^1 \left(\left[xy^2 \right]_{y=x^2}^{y=x} \right) dx \\
&= \frac{1}{2} \int_0^1 \left(x^3 - x^5 \right) dx \\
&= \frac{1}{2} \left[\frac{1}{4} x^4 - \frac{1}{6} x^6 \right]_0^1 \\
&= \frac{1}{24}
\end{aligned}$$

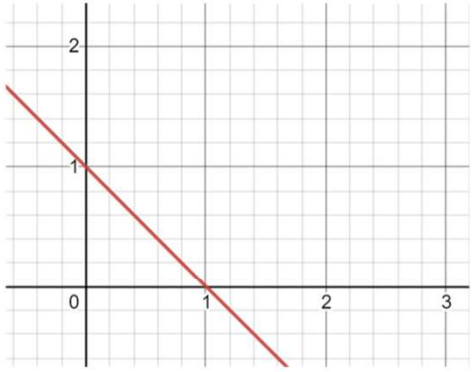


Or, $$\begin{aligned}
\iint_D xy\,dxdy &= \int_0^1 \left(\int_{x=y}^{x=\sqrt{y}} xy\,dx \right) dy \\
&= \frac{1}{2} \int_0^1 \left(\left[x^2 y \right]_{x=y}^{x=\sqrt{y}} \right) dy \\
&= \frac{1}{2} \int_0^1 \left(y^2 - x^3 \right) dx \\
&= \frac{1}{2} \left[\frac{1}{3} y^3 - \frac{1}{4} y^4 \right]_0^1 \\
&= \frac{1}{24}
\end{aligned}$$

b) Determine $\iint_D xy\,dxdy$, where D is the area in the first quadrant between the triangle with vertices $(0, 0)$, $(1, 0)$ and $(0, 1)$.

Solution. Here we have $0 \leq y \leq 1 - x$ for $0 \leq x \leq 1$.

$$\begin{aligned}
\iint_D xy\,dxdy &= \int_0^1 \left(\int_{y=0}^{y=1-x} xy\,dy \right) dx \\
&= \tfrac{1}{2} \int_0^1 \left(\left[xy^2 \right]_{y=0}^{y=1-x} \right) dx \\
&= \tfrac{1}{2} \int_0^1 x(1-x)^2\,dx \\
&= \tfrac{1}{2} \left[\tfrac{1}{2}x^2 - \tfrac{2}{3}x^3 + \tfrac{1}{4}x^4 \right]_0^1 \\
&= \tfrac{1}{24}
\end{aligned}$$



$y = 1 - x$

Note: It is a pure coincidence that the answer is the same as in part a).

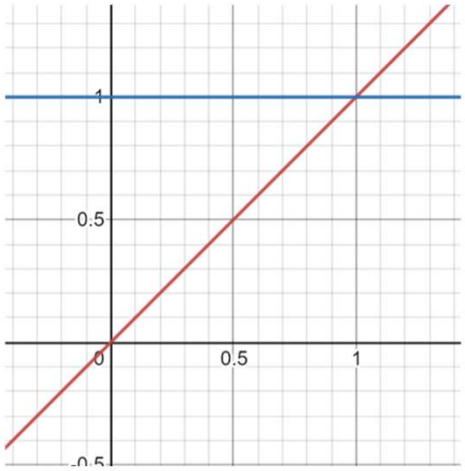
Interchanging the order of integration

We have the freedom to choose which double integral to start with, either dx first or dy, as $\iint_R f(x,y)dxdy = \iint_R f(x,y)dydx$. Of course, if we start the integration with dy first, then ensure that $\int \int_{y=} \Big(f(x,y)dy\Big)dx$ what is shown on the arrow are placed at the right functions.

$$\iint_D f(x,y)dxdy = \int_a^b \left[\int_{y=\alpha(x)}^{y=\beta(x)} f(x,y)\,dy\right]dx = \int_c^d \left[\int_{x=\gamma(y)}^{x=\delta(y)} f(x,y)\,dx\right]dy\,.$$

Example 8.10

Determine $\iint_T e^{-y^2}\,dxdy$, where T is the triangular region with vertices $(0, 0)$, $(1, 1)$ and $(0, 1)$.



We see that $x \leq y \leq 1$ in $0 \leq x \leq 1$.

Now consider $\int_0^1 \int_{y=x}^{y=1} e^{-y^2}\,dy\,dx$. We see that it is not easy to integrate (although one could use Taylor's series to integrate, try it). Therefore, we relook at another possibility and 'start' with the following:

We see that $0 \le x \le y$ in $0 \le y \le 1$.

$$\int_{y=0}^{y=1} \int_{x=0}^{x=y} e^{-y^2}\,dx\,dy = \int_{y=0}^{y=1} \left[xe^{-y^2} \right]_{x=0}^{x=y} dy = \int_{y=0}^{y=1} ye^{-y^2}\,dy = \left[-\tfrac{1}{2} e^{-y^2} \right]_0^1 = \tfrac{1}{2}(1 - e^{-1}).$$

Example 8.11

Evaluate $\int_{x=0}^{x=\pi/4} \int_{y=2x}^{y=\pi/2} \frac{\cos y}{y}\, dy\, dx$. We first note that it is not easy to find $\int_{y=2x}^{y=\pi/2} \frac{\cos y}{y}\, dy$. Take note $2x \le y \le \frac{\pi}{2}$ for $0 \le x \le \frac{\pi}{4}$; which is equivalent to $0 \le x \le \frac{y}{2}$ for $0 \le y \le \frac{\pi}{2}$.

$$\int_{y=0}^{y=\pi/2} \int_{x=0}^{x=y/2} \frac{\cos y}{y}\, dx\, dy$$

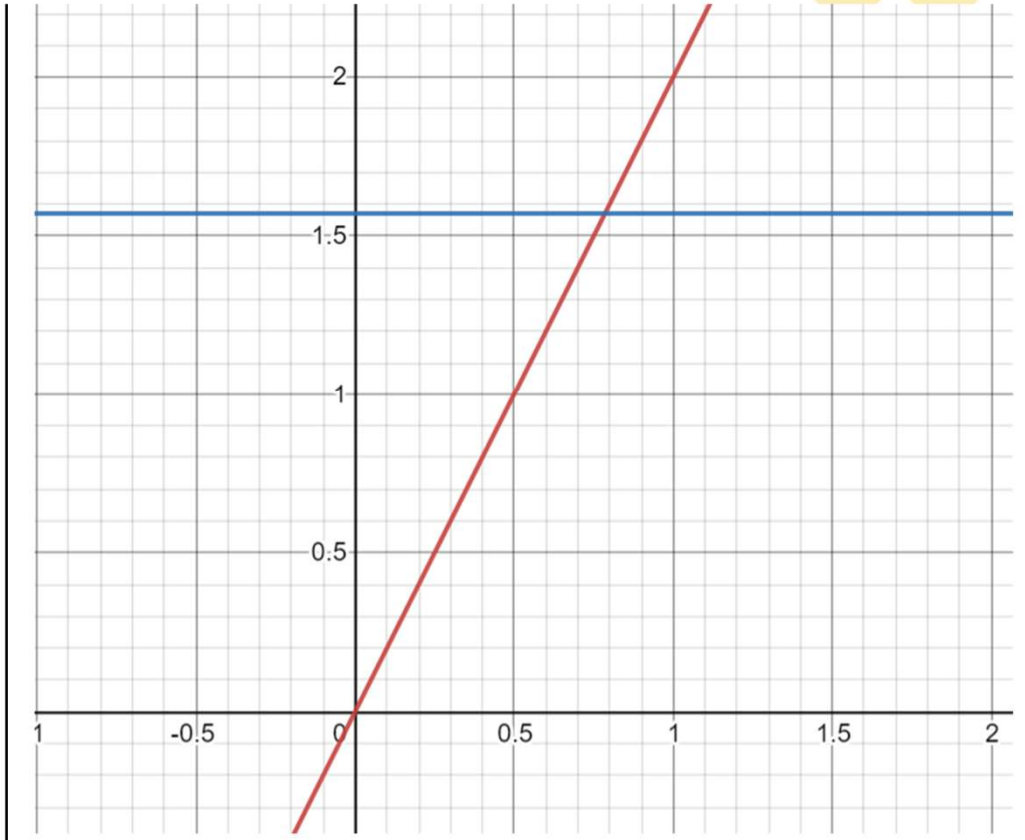
$$= \int_{y=0}^{y=\pi/2} \left[\frac{x \cos y}{y} \right]_{x=0}^{x=y/2} dy$$

$$= \frac{1}{2} \int_{y=0}^{y=\pi/2} \cos y\, dy$$

$$= \frac{1}{2} \left[\sin y \right]_{0}^{\pi/2}$$

$$= \frac{1}{2} \sin\left(\frac{\pi}{2}\right)$$

$$= \frac{1}{2}$$



Summary

1. The priority and the most important thing to do is to find the correct upper and lower limits based on the bounded region.
2. MUST draw the graph (like AFM MUST draw the timeline).
3. You have the freedom to start dx or dy first.
4. If start with dy then see horizontally bottom-up.
5. If start with dx then see vertically left-to-right.
6. Integrate carefully and avoid any careless mistakes.
7. If time allows, do on both double integral to confirm the answer.

Apply double integration to find the area of a region

Consider a region D. The precise area is obtained by

$$\text{Area of } D = \iint_D 1\, dA = \iint_D 1\, dx dy\,.$$

Example 8.12

Find the area of the region inside the unit circle $x^2 + y^2 = 1$.

$$\text{Area} = 2 \times \int_{x=-1}^{x=1} \int_{y=0}^{y=\sqrt{1-x^2}} 1 \, dy \, dx = 2 \int_{x=-1}^{x=1} \left[y \right]_{y=0}^{y=\sqrt{1-x^2}} dx = 2 \int_{x=-1}^{x=1} \sqrt{1-x^2} \, dx .$$

Let $x = \sin\theta \Rightarrow \sqrt{1-x^2} = \cos\theta$ and $dx = \cos\theta d\theta$

$$\Rightarrow \int_{-1}^{1} \sqrt{1-x^2} dx = \int_{-1}^{1} \cos^2\theta \, d\theta = \frac{1}{2} \int_{\sin^{-1}(-1)}^{\sin^{-1}1} (1 + \cos 2\theta) \, d\theta = \left[\frac{1}{2}\theta + \frac{1}{4}\sin 2\theta \right]_{-\pi/2}^{\pi/2} = \frac{\pi}{2} .$$

Hence, the area is $2 \times \frac{\pi}{2} = \pi$.

Definition 8.13: Jacobian

Definition. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ such that $f(u,v) = (f_1(u,v), f_2(u,v)) = (x,y)$ (here we write $f_1(u,v) = x(u,v), f_2(u,v) = y(u,v)$). A Jacobian of x (or f_1) and y (or f_2) with respect to u and v (is denoted as J), is the determinant of the total derivative of f,

$$J = \det\begin{pmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_1}{\partial v} \\ \frac{\partial f_2}{\partial u} & \frac{\partial f_2}{\partial v} \end{pmatrix} = \det\begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}$$

Now we show how to apply the substitution method in double integral:

1. Substitute suitable new functions x and y in term of both u and v.
2. Then substitute/change dx into du and dy into dv, <u>but</u> there is something more next.
3. To multiply $\left|J\right|$, the absolute value of Jacobian. We do this because the choice of x and y, and u and v are arbitrary. So, the absolute value ensures there is no difference at the outcome.
4. In addition, if we deal with definite integral, do remember to make necessary changes.

Theorem 8.14

$$\text{In short, } \iint\limits_{D} f(x,y)\,dxdy = \iint\limits_{D'} f(x(u,v),y(u,v))\left|\det\begin{pmatrix}\frac{\partial x}{\partial u} & \frac{\partial x}{\partial v}\\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v}\end{pmatrix}\right| du dv\,.$$

Example 8.15: Polar coordinates substitution

When we substitute of $x = r\cos\theta$ and $y = r\sin\theta$, then

$$J = \det\begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \det\begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix} = r\,.$$

(Make sure the upper/lower limits are at least 0)

Hence, $$\boxed{\iint_D f(x,y)\,dxdy = \iint_{D'} \Big(f(x(r,\theta),y(r,\theta))\cdot r\Big)\,drd\theta\,.}$$

(Remember to multiply r)

Example 8.16: Area of a circle with radius *a*

By the substitution method:

Step 1. Let $x = r\cos\theta$ and $y = r\sin\theta$ where $0 \le \theta \le 2\pi, 0 \le r \le a$

Step 2. Then $dxdy = \left|J\right| drd\theta$.

Step 3. $J = \det\begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \det\begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix} = r$.

$$\text{Area} = \iint_D 1\, dxdy = \int_{\theta=0}^{\theta=2\pi}\int_{r=0}^{r=a} r\, dr\, d\theta = \tfrac{1}{2}\int_{\theta=0}^{\theta=2\pi}\left[r^2\right]_{r=0}^{r=a} d\theta = \tfrac{1}{2}\int_0^{2\pi} a^2\, d\theta = \tfrac{1}{2}a^2\theta\Big|_0^{2\pi} = \pi a^2$$

Example 8.17(a)

Compute $\iint\limits_D e^{-(x^2+y^2)}\,dxdy$ with the integration region

$D = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \le a^2\}.$

Solution. Let $x = r\cos\theta$ and $y = r\sin\theta$ with $0 \le \theta \le 2\pi, 0 \le r \le a$. Then

$$\iint\limits_D e^{-(x^2+y^2)}\,dxdy = \int\limits_{\theta=0}^{\theta=2\pi}\int\limits_{r=0}^{r=a} e^{-r^2}\,r\,dr\,d\theta = -\tfrac{1}{2}\int\limits_{\theta=0}^{\theta=2\pi}\left[e^{-r^2}\right]_{r=0}^{a} d\theta$$

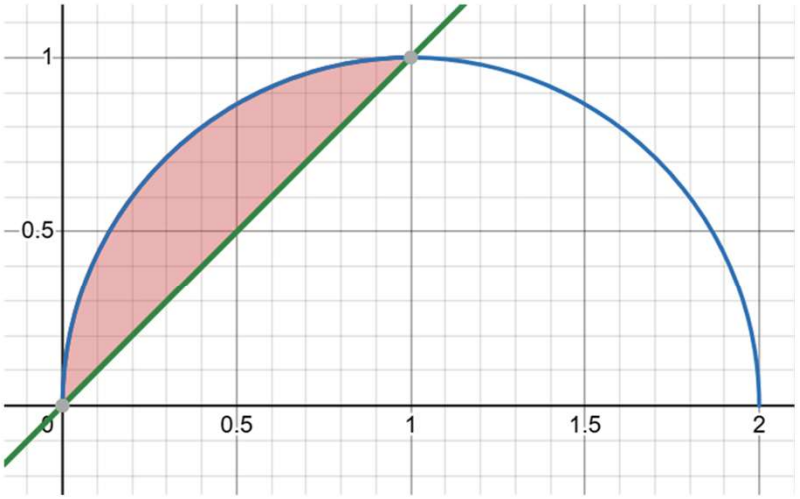
$$= \tfrac{1}{2}\int\limits_0^{2\pi}(1-e^{-a^2})\,d\theta = \tfrac{1}{2}\theta(1-e^{-a^2})\Big|_0^{2\pi} = \pi(1-e^{-a^2})$$

Example 8.17(b)

Sketch the graph of $x \leq y \leq \sqrt{2x - x^2}$ and $y = x$ for $0 \leq x \leq 1$ and then evaluate $I = \int_0^1 \int_x^{\sqrt{2x-x^2}} \sqrt{4 - x^2 - y^2}\, dydx$ using the polar coordinates substitution.

Note: $\int \sin^3 u \, du = \int \sin u(1 - \cos^2 u) \, du = -\cos u + \frac{1}{3}\cos^3 u + C$

Solution. Note that $y = \sqrt{2x - x^2} \Rightarrow (x-1)^2 + y^2 = 1$ for $y \geq 0$.



The semi-circle has the equation:

$x^2 + y^2 = 2x \Rightarrow r^2 = 2r\cos\theta$

$\Rightarrow r = 2\cos\theta, r \neq 0$

Let $x = r\cos\theta, y = r\sin\theta$.

For $0 \leq r \leq 2\cos\theta$ when $\frac{\pi}{4} \leq \theta \leq \frac{\pi}{2}$:

$$I = \int_0^1 \int_x^{\sqrt{2x-x^2}} \sqrt{4 - x^2 - y^2}\, dydx$$

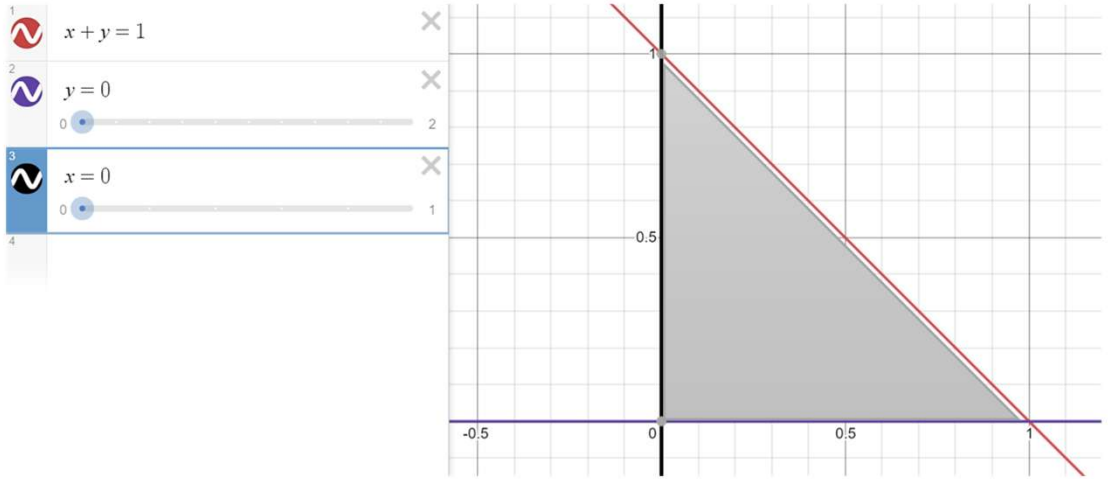
$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \int_0^{2\cos\theta} \sqrt{4 - r^2}\, rdrd\theta = -\frac{1}{3}\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \left[(4 - r^2)^{\frac{3}{2}}\right]_0^{2\cos\theta} d\theta$$

$$= \frac{8}{3}\int_{\frac{\pi}{\pi}}^{\frac{\pi}{2}} (1 - \sin^3\theta)d\theta = \frac{8}{3}\left(\frac{\pi}{4} - \frac{5}{6\sqrt{2}}\right)$$

Example 8.18: Special substitution

Evaluate $I = \iint_D \left(\frac{x-y}{x+y}\right)^2 dydx = \int_0^1 \int_0^{1-x} \left(\frac{x-y}{x+y}\right)^2 dydx$.

Solution. We sketch the graph for $0 \leq y \leq 1 - x$ on $0 \leq x \leq 1$.



Since the integration is not easy to integrate, we need to convert this into u/v so that can be done easier. This prompt us to make substitution of

$$u = x + y, v = x - y \Rightarrow x = \tfrac{1}{2}(u + v), y = \tfrac{1}{2}(u - v)\,.$$

We need to find the Jacobian so that we could replace $dydx = |J\,|dudv$.

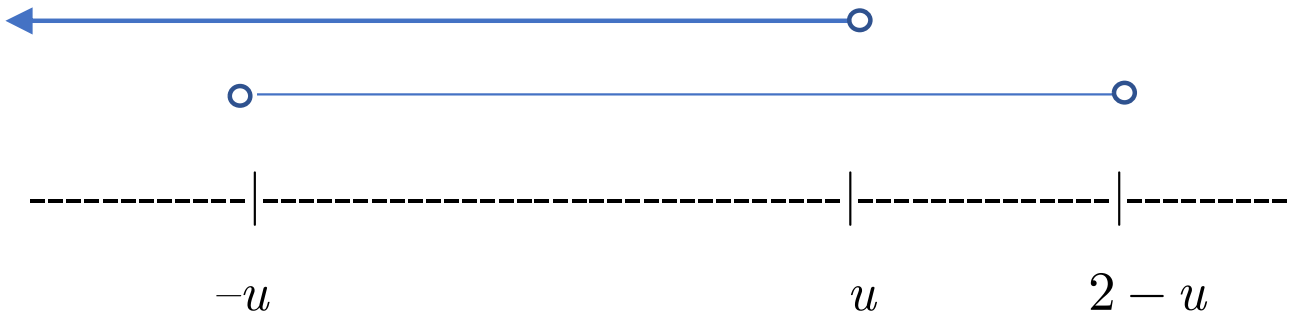
$$J = \det\begin{pmatrix} \frac{dx}{du} & \frac{dx}{dv} \\ \frac{dy}{du} & \frac{dy}{dv} \end{pmatrix} = \det\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix} = -\tfrac{1}{4} - \tfrac{1}{4} = -\tfrac{1}{2} \Rightarrow \left|J\right| = \tfrac{1}{2}\,.$$

We also need to convert $0 \leq y \leq 1 - x$ and $0 \leq x \leq 1$ to intervals of u and v. First, we see that the shaded area indicated that $0 \leq u = x + y \leq 1$.

$$0 \leq x = \tfrac{1}{2}(u + v) \leq 1 \Rightarrow 0 \leq u + v \leq 2 \Rightarrow -u \leq v \leq 2 - u \qquad (1)$$

$$0 \leq y \leq 1 - x \Rightarrow 0 \leq \tfrac{1}{2}(u - v) \leq 1 - \tfrac{1}{2}(u + v) \Rightarrow v \leq u \leq 1 \qquad (2)$$

To fulfill (1) and (2), we find the interval for v:



Note that $2-u \geq u$ and we could prove this using the method of 'contradiction proof'. Assume the contrary, where $2-u < u$ then we will obtain $u > 1$. But this is against the fact that $u \leq 1$.

$$I = \int_0^1 \int_{-u}^{u} \left(\frac{v}{u}\right)^2 \cdot \frac{1}{2}\, dv du = \frac{1}{6} \int_0^1 \frac{v^3}{u^2}\Big|_{-u}^{u} du = \frac{1}{3} \int_0^1 u\, du = \frac{1}{6}.$$

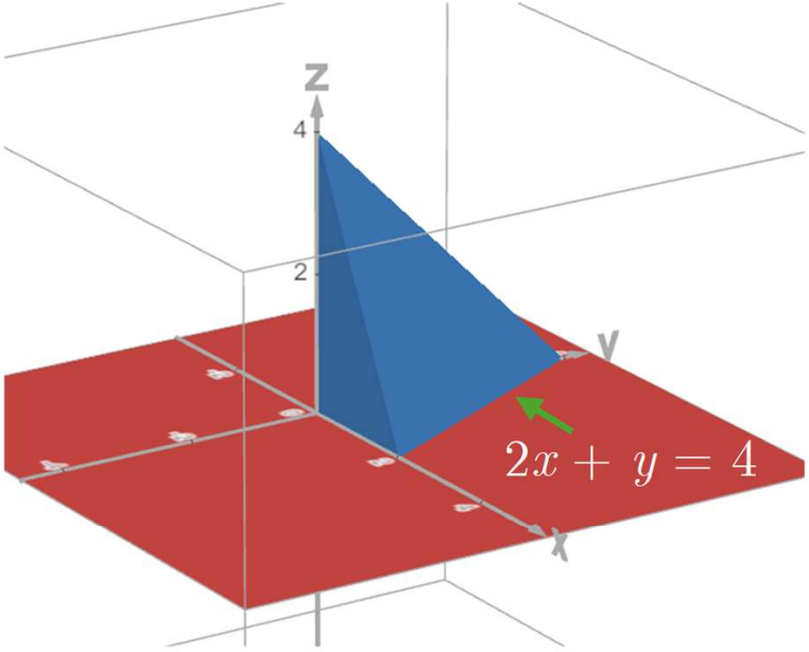
Apply double integral to find volume

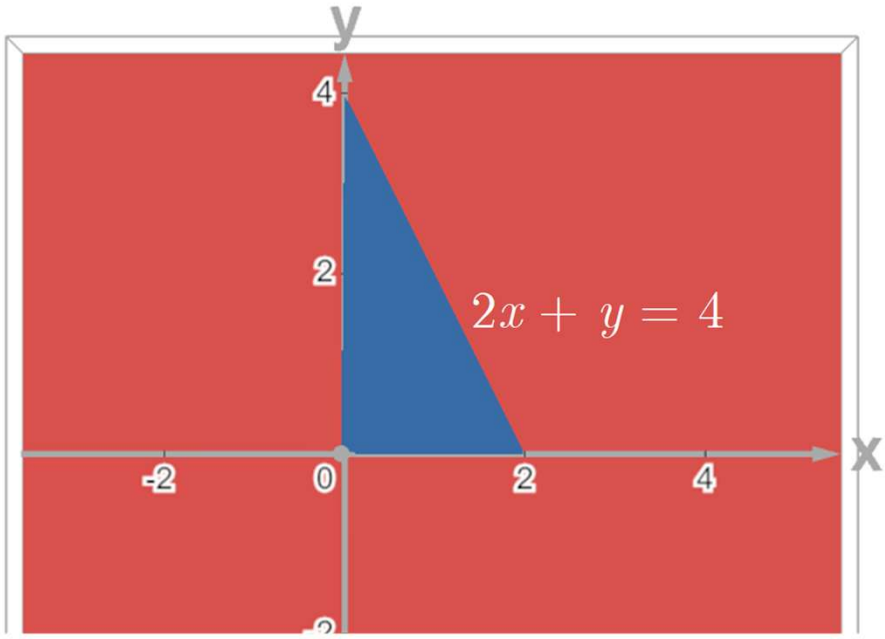
Volume generated by the region D from $z = 0$ going upward to $z = f(x, y)$ is given by

$$\boxed{\iint_D f(x, y)dxdy}$$

Example 8.19

Find the volume bounded by plane $z = 4 - 2x - y$ and the coordinates planes.





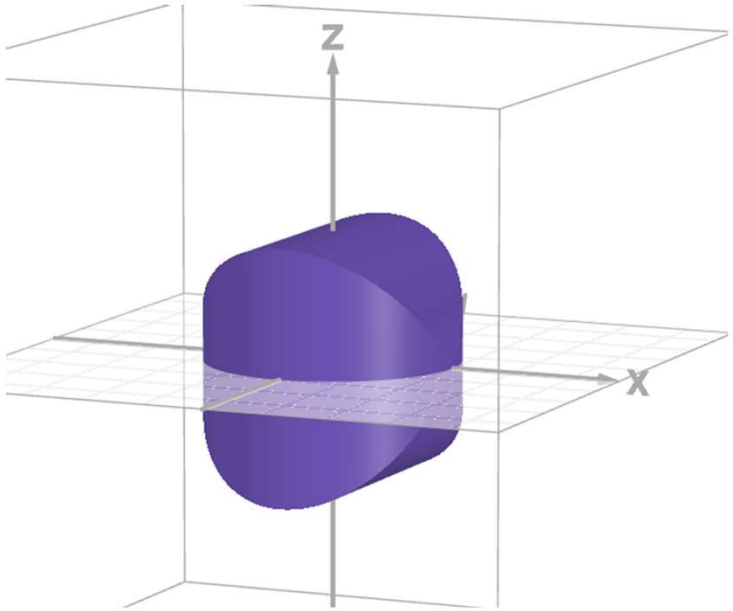
$$V = \int_0^4 \int_0^{\frac{1}{2}(4-y)} (4 - 2x - y)\, dxdy = \frac{16}{3} \text{ or } V = \int_0^2 \int_0^{4-2x} (4 - 2x - y)\, dydx = \frac{16}{3}.$$

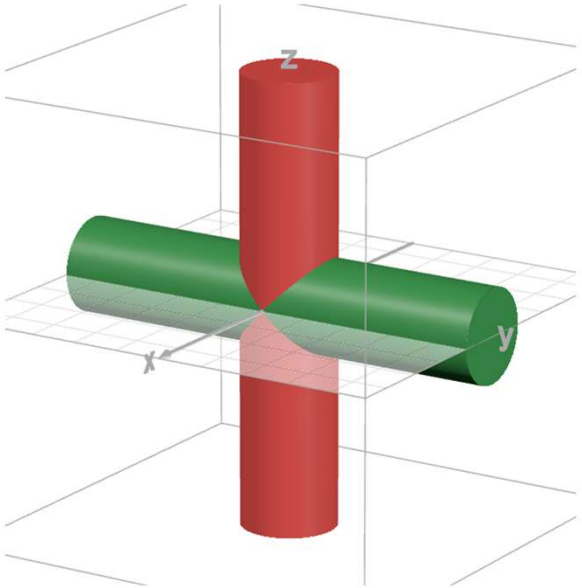
Example 8.20

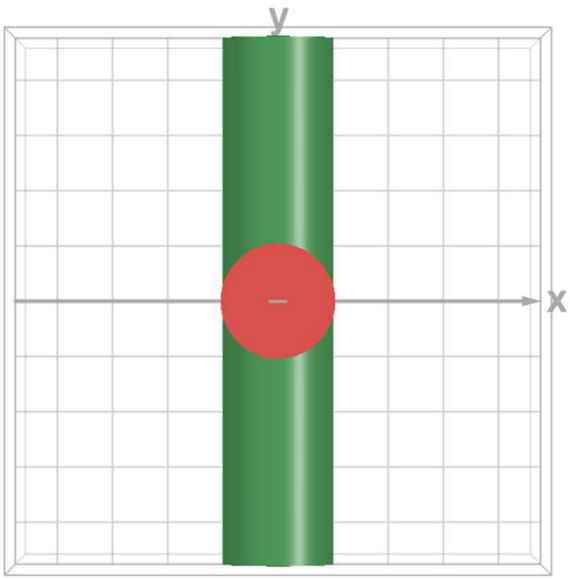
Find the volume of the region inside the intersection of two cylinders:

$x^2 + y^2 = a^2$ and $x^2 + z^2 = a^2$.

Solution. Note that D is a circle centred at the origin with radius a.







Consider the first octant and we multiply the answer by eight:

$$V = 8\int_0^a \int_0^{\sqrt{a^2-x^2}} z\,dydx = 8\int_0^a \int_0^{\sqrt{a^2-x^2}} \sqrt{a^2-x^2}\,dydx = 8\int_0^a \left[y\sqrt{a^2-x^2}\right]_0^{\sqrt{a^2-x^2}} dx$$

$$= 8\int_0^a (a^2-x^2)\,dx = 8\left[a^2x - \tfrac{1}{3}x^3\right]_0^a = \tfrac{16}{3}a^3$$

Alternative:

$$V = 8\int_0^{\frac{\pi}{2}}\int_0^{a}\sqrt{a^2 - r^2\cos^2\theta}\; r\,dr\,d\theta = -\frac{8}{3}\int_0^{\frac{\pi}{2}}\left[\frac{(a^2 - r^2\cos^2\theta)^{3/2}}{\cos^2\theta}\right]_0^{a} d\theta = -\frac{8}{3}a^3\int_0^{\frac{\pi}{2}}\frac{(1-\cos^2\theta)^{3/2}-1}{\cos^2\theta}\; d\theta$$

$$= -\frac{8}{3}a^3\int_0^{\frac{\pi}{2}}\frac{\sin^3\theta - 1}{\cos^2\theta}\; d\theta = -\frac{8}{3}a^3\int_0^{\frac{\pi}{2}}\left(\frac{1-\cos^2\theta}{\cos\theta}\tan\theta - \sec^2\theta\right) d\theta$$

$$= -\frac{8}{3}a^3\int_0^{\frac{\pi}{2}}\left(\sec\theta\tan\theta - \sin\theta - \sec^2\theta\right) d\theta = -\frac{8}{3}a^3\left[\sec\theta + \cos\theta - \tan\theta\right]_0^{\frac{\pi}{2}}$$

$$= -\frac{8}{3}a^3\left[\frac{1-\sin\theta}{\cos\theta} + \cos\theta\right]_0^{\frac{\pi}{2}} = -\frac{8}{3}a^3\left(\lim_{b\to\frac{\pi}{2}}\frac{1-\sin b}{\cos b} - 2\right) = -\frac{8}{3}a^3\left(\lim_{b\to\frac{\pi}{2}}\frac{-\cos b}{-\sin b} - 2\right) = \frac{16}{3}a^3$$